VEMULAPALLI AKSHAY

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SUMMARY

CS & Mathematics undergraduate at NUS (GPA 4.75/5.0, Dean's List) who writes problems that frontier AI models fail. Paid contributor on **Shipd**, authoring ML, data-science and software-engineering challenges with reference solutions and test suites for model evaluation. Experienced in **reviewing and grading code at scale** (CS1101S TA, Source Academy autograder backend), **agentic AI workflows** with Claude Code, and LLM agent memory systems. Strong in Python, C++, TypeScript and mathematical reasoning.

SKILLS

AI / LLM: LLM evaluation & adversarial task design, rubric-based grading, Claude Code (agentic workflows, custom skills, automation), prompt engineering, RAG & agent memory architectures

Languages: Python, C++17, C, TypeScript, JavaScript, Java, SQL (SQLite / MySQL)

Frameworks & Tools: React.js, Node.js, Boost.Beast, Google Test, Docker, Git / GitHub, CI/CD, CMake, Azure, AWS Lambda, Linux (Arch, Ubuntu)

Strengths: Code review & debugging, test-case and edge-case design, concurrency, algorithms & complexity analysis, clear technical writing

EXPERIENCE

Shipd — *Contributor, AI Training & Evaluation (Freelance, Remote)* Sep 2026 – Present

- **Author challenge problems built to break frontier AI models** across ML model evaluation, data science, algorithms and software engineering, each with a reference solution and test suite
- **Submissions pass a multi-stage expert review** for correctness, difficulty and clarity; paid per accepted challenge
- **Built a Claude Code–driven pipeline** (custom skills and automations) to draft, validate and stress-test submissions against models before review

National University of Singapore — *Teaching Assistant, CS1101S Programming Methodology* Aug 2026 – Present

- **Lead weekly studios and assess first-year students' code** on recursion, functional programming and computational models (SICP-Py curriculum), giving written, rubric-based feedback

Source Academy — NUS School of Computing — *Head of Backend Systems* Aug 2026 – Present

- **Own the backend server processes and autograder Lambdas** that grade programming assignments for the CS1101S cohort; maintain pipeline integrations and resolve module issues

Software Developer (4-credit independent work module) Jan 2026 – Apr 2026

- **Won Best Project – Platinum, 28th NUS STePS** (highest distinction) for CSE Machine visualisation; implemented liveness analysis and a "Clear Dead Frames" toggle; fixed React / TypeScript UI bugs for 500+ concurrent users (merged PRs)

NUS Teaching Module Task Force — CS1101S Language Migration — *Student Software Engineer* May 2026 – Present

- **Migrating NUS's intro CS course** from JavaScript-based Source to Python, and from the legacy Immer framework to Conductor, for all incoming CS undergraduates; built the CSE Machine visualiser plugin and its runner-side plugin

PROJECTS

Generative-Agent Memory System — *Personal Project | Python, LLMs* 2025 – 2026

- **Rebuilt a Smallville-style (Generative Agents) memory architecture** for multi-character LLM agents: each agent keeps its own memory stream, personality and key facts, retrieved per query by recency, importance and relevance
- **Gave more consistent long-running, multi-agent conversations** than flat RAG or a shared memory store, keeping each character faithful within one session

O(1) Exchange — Orbital 2026 (Apollo) — *Systems & Full-stack Engineer | C++17, Docker, Azure* May 2026 – Present

- **Built a sub-millisecond C++17 order-matching engine** (price-time priority) with a Boost.Beast WebSocket server for concurrent multi-client sessions; per-instrument mutex locking
- **Engineered auth and portfolio layers** (libsodium Argon2id, session tokens, SQLite3); deployed on Azure with Docker, CI/CD and a Google Test suite

EDUCATION

National University of Singapore — *B.Comp in Computer Science and Mathematics* Aug 2025 – Present

GPA: 4.75 / 5.0 | Dean's List AY25/26 Sem 2 (SGPA 4.875, top 5% of CS cohort) | Invite-only Independent SWE Work module

IIIT Hyderabad — *B.Tech CS & M.Sc Computational Science (transferred to NUS)* Aug 2024 – Mar 2025

GPA: 9.8 / 10.0 | Data Structures & Algorithms, Software Systems, Digital Systems

Sahyadri School KFI — *ISC (Grade 12) & ICSE (Grade 10)* Jun 2017 – Mar 2024

ISC: 97.8% (top 1% in India), Mathematics 100/100 (top 0.1% nationally) | ICSE: 96.5% | Strong JEE Advanced rank

OTHER

- Terminal multiplayer game (team lead, 24-hour hackathon): real-time 2-player game over TCP in C | RC4 Inter-College Debate Representative